



IEETA



Exploring the Effect of Personalized Virtual Reality Serious Game for Stroke Rehabilitation

MAPDE 2025/2026 – Investigation Plan Presentation

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INTRODUCTION

Why this Project?

- How does **stroke** impacts our society?
- Why **VR** for rehabilitation?



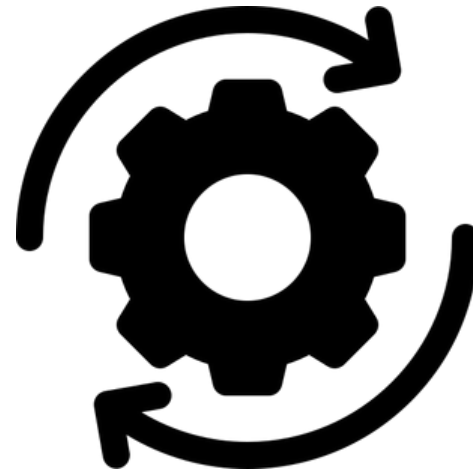
World Stroke
Organization



GOALS



**Develop a Functional
Game Prototype**



**Implement algorithms/rules
for dynamic difficulty
adjustment based on user
performance.**



**Test the prototype with
users**

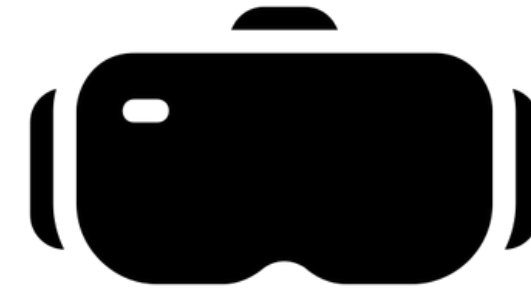
BENEFITS



**Increase survivors'
motivation**

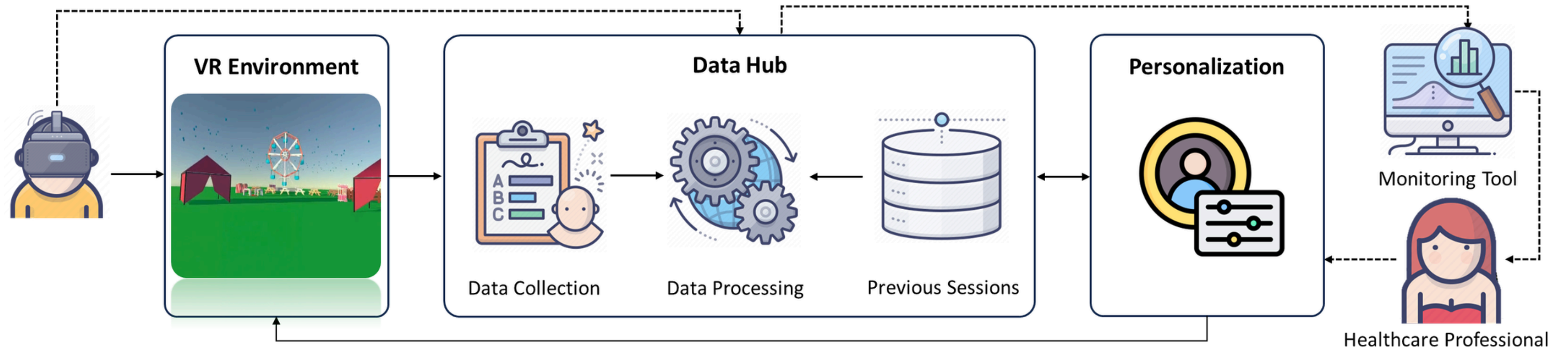


**Improves survivors'
progress**



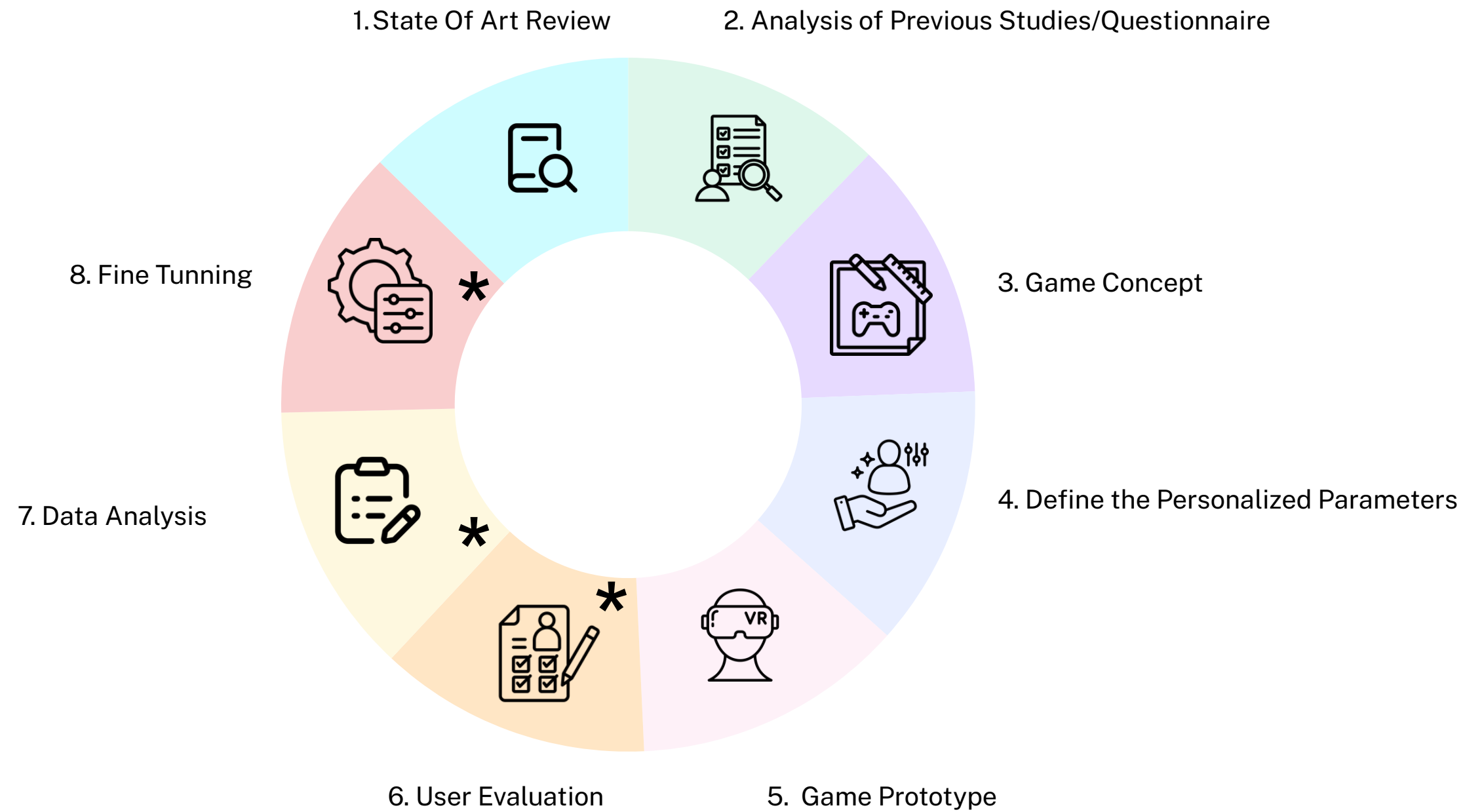
**Contribute to VR
Rehabilitation
Research**

CONCEPTUAL FRAMEWORK

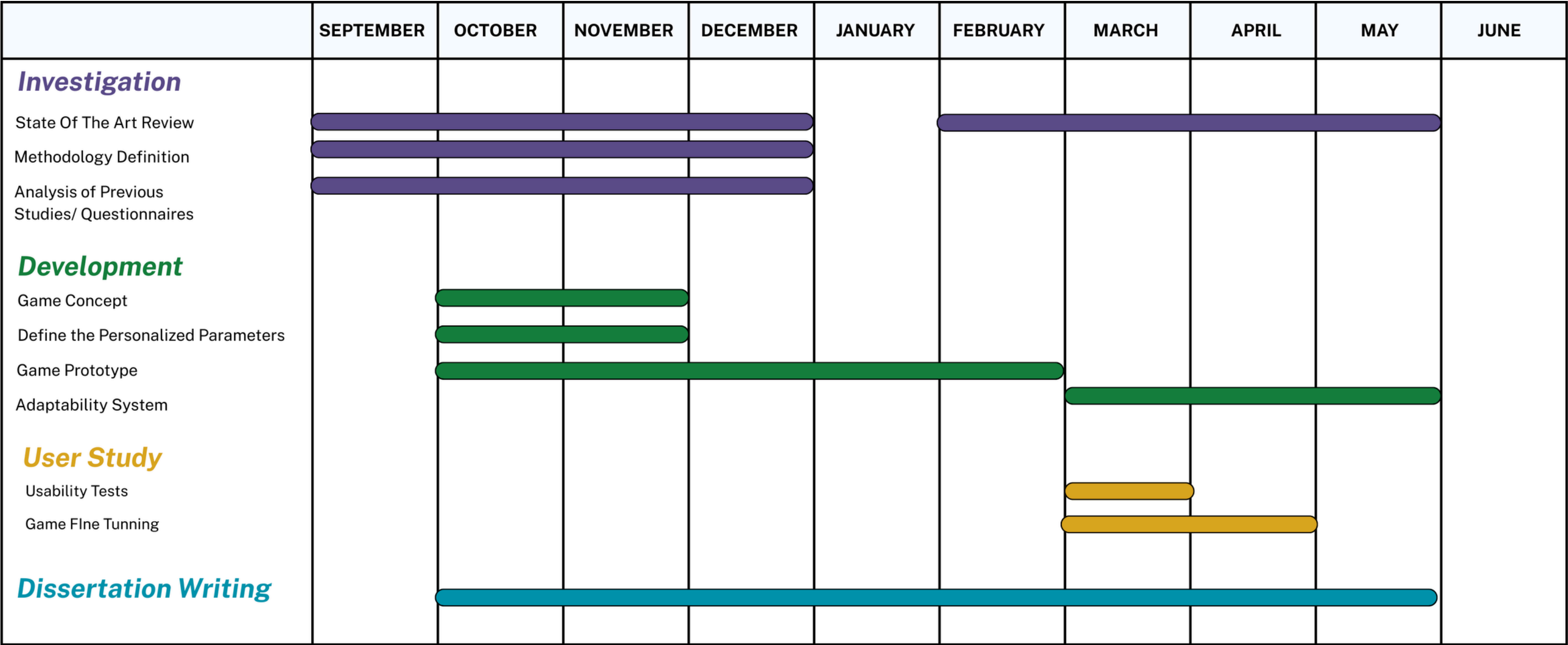


METHODOLOGY

- This project will adopt the Human-Centered Design (HCD)



TIMELINE



EXPECTED RESULTS

Answer the Research Question

- Can VR based environments be personalized to adjust to individual need and help boost motivation?

At a development level

- Have different variations of the functional prototype to test different approaches of dynamic difficulty adjustment

At a therapeutically level

- Reinforce the use of serious games with virtual reality environments as as complementary methods to the traditional ones to boost survivors motivation and engagement

The End

Thank You For Listening